

IMO 2026 — Autonomous Solutions

GPT-5.6 Sol

max reasoning effort (base model)

Independently graded: 30 / 42

P1 7 P2 7 P3 1 P4 7 P5 7 P6 1

All six problems attempted fully autonomously in one identical minimal agent loop (bash / write_file / read_file), network blocked, 150 min per-problem cap. Scores are IMO-style 0–7 from independent verifier agents that re-derived key algebra symbolically and machine-tested claims — not the model’s self-reports. Per-turn audit trails, grader verdicts, and run statistics accompany this document in the `imo-2026` repository.

IMO 2026 — Problem 1

Solution by GPT-5.6 Sol (max reasoning)

2026-07-22

Source: `results/gpt-5.6-sol-max/problem-01/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\text{lcm}(m, n)}{\gcd(m, n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

Solution document

Status

solved

Problem

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Approaches tried

- **Lexicographic descent:** Tracking the product P of all entries and the number r of entries greater than 1 gives a lexicographically decreasing pair (P, r) . This was consolidated into the single positive-integer potential $P2^r$.
- **Prime-adic exponent invariant:** For each prime p , a selected pair of exponents (a, b) changes to $(\min(a, b), |a - b|)$. The gcd of all p -adic exponents is invariant and determines the exponent of p in the terminal number.

- **Exponent-sum exploration:** The sum of the p -adic exponents is nonincreasing, but it can stay unchanged when one selected entry is not divisible by p , so it is not by itself a termination argument. It nevertheless helps motivate separating termination from the choice-independent invariant.
- **Computational sanity check:** Exhaustive exploration of all move trees for 705 small initial multisets (sizes 2 through 4, entries at most 10) confirmed strict potential descent and the stated terminal formula.

Current best

The problem is solved. If the initial entries are $x_1, \dots, x_{2026}$, then every play terminates with one nonunit, and its choice-independent value is

$$M = \prod_p p^{\gcd(v_p(x_1), \dots, v_p(x_{2026}))},$$

where the gcd of an all-zero list is 0; only finitely many factors differ from 1.

Full proof

Let $N = 2026$. At any stage, denote the entries on the board by

$$y_1, y_2, \dots, y_N,$$

and define

$$P = \prod_{i=1}^N y_i, \quad r = \#\{i : y_i > 1\}, \quad \Phi = P2^r.$$

Thus Φ is a positive integer.

(a) Termination and the number of remaining nonunits

Suppose a legal move selects $m, n > 1$, and write

$$d = \gcd(m, n).$$

Using $\gcd(m, n) \operatorname{lcm}(m, n) = mn$, the product of the two replacement entries is

$$d \cdot \frac{\operatorname{lcm}(m, n)}{d} = \operatorname{lcm}(m, n) = \frac{mn}{d}.$$

Consequently, if P' denotes the product after the move, then

$$P' = \frac{P}{d}.$$

If $d = 1$, then the replacement pair is $(1, mn)$. Hence two nonunits are replaced by exactly one nonunit, so $r' = r - 1$. Therefore

$$\Phi' = P'2^{r'} = P2^{r-1} = \frac{\Phi}{2} < \Phi.$$

If $d > 1$, then the first replacement entry d is greater than 1, while the second is a positive integer. Thus the two selected nonunits are replaced by either one or two nonunits, and in particular $r' \leq r$. It follows that

$$\Phi' = \frac{P}{d} 2^{r'} \leq \frac{P}{d} 2^r = \frac{\Phi}{d} < \Phi.$$

Hence every legal move strictly decreases the positive integer Φ . There is no infinite strictly decreasing sequence of positive integers, so every sequence of moves is finite.

When no further move is possible, there cannot be two entries greater than 1, because any such two entries could be selected. Thus a terminal board has at most one nonunit. On the other hand, a move can never eliminate all nonunits: indeed, the product of its two replacement entries is $\text{lcm}(m, n) > 1$, so those entries cannot both be 1. Since initially all entries are greater than 1, at least one nonunit remains at every stage. Therefore, when the process terminates, exactly one entry M is greater than 1.

(b) Independence of the choices

Let the initial entries be $x_1, \dots, x_N$. Fix a prime p . For a current board $y_1, \dots, y_N$, define

$$G_p = \gcd(v_p(y_1), \dots, v_p(y_N)),$$

with the convention that the gcd of an all-zero list is 0.

Suppose the two entries selected in a move have p -adic exponents

$$a = v_p(m), \quad b = v_p(n).$$

The two replacement entries have exponents

$$v_p(\gcd(m, n)) = \min(a, b)$$

and

$$v_p\left(\frac{\text{lcm}(m, n)}{\gcd(m, n)}\right) = \max(a, b) - \min(a, b) = |a - b|.$$

Furthermore,

$$\gcd(\min(a, b), |a - b|) = \gcd(a, b).$$

To verify this identity, if $a \geq b$, an integer divides both a and b if and only if it divides both b and $a - b$; the case $b \geq a$ is symmetric. This also covers the cases in which one or both exponents are zero.

Thus the gcd of the two altered exponents does not change. Taking its gcd with all the unaltered exponents shows that G_p is invariant under every move.

At the terminal board, all entries except M equal 1. Since $v_p(1) = 0$, the terminal value of this invariant is

$$\gcd(0, \dots, 0, v_p(M), 0, \dots, 0) = v_p(M).$$

It must equal its initial value, so for every prime p ,

$$v_p(M) = \gcd(v_p(x_1), \dots, v_p(x_N)).$$

Therefore

$$M = \prod_p p^{\gcd(v_p(x_1), \dots, v_p(x_N))}.$$

For every prime not dividing any initial entry, the exponent in this product is 0, so the product is finite. The right-hand side depends only on the initial board, and uniqueness of prime factorization now proves that the terminal value M is independent of every choice made during the process. $\square$

IMO 2026 — Problem 2

Solution by GPT-5.6 Sol (max reasoning)

2026-07-22

Source: `results/gpt-5.6-sol-max/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC , respectively. Let points K and L be chosen inside triangles BMC and BNC , respectively, such that K lies inside the angle LBA , L lies inside the angle ACK , and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL . Prove that $OM = ON$.

Solution document

Status

solved

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC , respectively. Let points K and L be chosen inside triangles BMC and BNC , respectively, such that K lies inside the angle LBA , L lies inside the angle ACK , and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL . Prove that $OM = ON$.

Approaches tried

- **Power/circle-equation reformulation:** rewrote $OM = ON$ as a linear relation between the coefficients of the circumcircle of AKL .
- **Angle parametrization and sine law:** introduced $\alpha, \beta, \gamma, x, y, z$ and derived exact sine-law expressions for BK and CL .
- **Normalized-length quadratics:** set $R = 2BK/AB$, $D = 2CL/AC$ and eliminated y, z , obtaining two compact quadratic relations through a common polynomial F .
- **Oblique coordinates/full elimination:** verified the identity but initially produced unwieldy algebra; this route also exposed and corrected one false intermediate symbolic claim.

- **Cartesian circle computation:** used simple ray-direction coordinates for K, L . Sequential exact polynomial division yielded a sparse certificate expressing the target in terms of the two geometric quadratic relations and the identities $\sin^2 + \cos^2 = 1$.
- **Synthetic/power exploration:** considered equal powers and inversion, but no comparably clear synthetic construction was found.

Current best

A complete analytic proof has been obtained. With $S = \sin \alpha, C = \cos \alpha, U = \sin x, X = \cos x, V = \sin(\alpha + x), R = 2BK/AB$, and $D = 2CL/AC$, the angle conditions imply

$$2U \sin \beta = \sin \gamma F(R), \quad 2U \sin \gamma = \sin \beta F(D), \quad F(T) = V(T^2 - 2XT + 2) - ST.$$

A verified explicit polynomial identity then gives

$$b(CP + SQ) - cP = \frac{b^2 - c^2}{2}$$

for the Cartesian coefficients of the circumcircle of AKL , which is exactly $OM^2 - ON^2 = 0$. The sparse certificate and its scaling were independently checked by `code/verify_sparse_identity.py`; the ray orders, positivity of every divisor, coordinate directions, and all circle formulas have also been audited in `scratch/audit.md`.

Full proof

Let

$$\alpha = \angle BAC, \quad \beta = \angle ABC, \quad \gamma = \angle ACB,$$

and set

$$x = \angle KBA = \angle ACL, \quad y = \angle LBK = \angle LNC, \quad z = \angle LCK = \angle BMK.$$

Also write

$$a = BC, \quad b = CA, \quad c = AB.$$

Because K lies inside $\angle LBA$, the rays at B occur in the order

$$BA, BK, BL, BC;$$

because L lies inside $\angle ACK$, the rays at C occur in the order

$$CA, CL, CK, CB.$$

In particular, all angles and sines used below have the indicated positive values.

1. Relations forced by the angle conditions

In $\triangle BMK$ we have $BM = c/2$, $\angle MBK = x$, and $\angle BMK = z$. Hence

$$BK = \frac{c}{2} \frac{\sin z}{\sin(x+z)}. \quad (1)$$

Moreover,

$$\angle KBC = \beta - x, \quad \angle BCK = \gamma - x - z,$$

so $\angle BKC = \alpha + 2x + z$. Thus

$$BK = a \frac{\sin(\gamma - x - z)}{\sin(\alpha + 2x + z)}. \quad (2)$$

Similarly, sine rule in $\triangle CNL$ and $\triangle BCL$ gives

$$CL = \frac{b}{2} \frac{\sin y}{\sin(x + y)} = a \frac{\sin(\beta - x - y)}{\sin(\alpha + 2x + y)}. \quad (3)$$

Normalize the circumdiameter of ABC to 1. Then

$$a = \sin \alpha, \quad b = \sin \beta, \quad c = \sin \gamma. \quad (4)$$

Put

$$S = \sin \alpha, \quad C = \cos \alpha, \quad U = \sin x, \quad X = \cos x, \quad V = \sin(\alpha + x) = SX + CU, \quad (5)$$

and

$$R = \frac{2BK}{c}, \quad D = \frac{2CL}{b}. \quad (6)$$

Thus

$$R = \frac{\sin z}{\sin(x + z)}, \quad D = \frac{\sin y}{\sin(x + y)}. \quad (7)$$

Let (θ, t, ρ) denote either (γ, z, R) or (β, y, D) . Equating the appropriate expressions in (1)–(3) and using (4) gives

$$\sin \theta \sin t \sin(\alpha + 2x + t) = 2S \sin(x + t) \sin(\theta - x - t). \quad (8)$$

Since $t > 0$, divide by $\sin^2 t$. From $\rho = \sin t / \sin(x + t)$,

$$\cot t = \frac{1/\rho - X}{U}. \quad (9)$$

Writing all terms in (8) in terms of $\cot t$ and using (9) yields

$$-\rho^2 \sin \theta U \sin(\alpha + x) + \rho \sin \theta (\cos(\alpha - x) - X \cos(\alpha + 2x)) - 2SU \sin(\theta - x) = 0. \quad (10)$$

We use

$$\cos(\alpha - x) - X \cos(\alpha + 2x) = U(2CUX - 2SU^2 + 3S), \quad (11)$$

and

$$\cot \gamma = \frac{b - Cc}{Sc}, \quad \cot \beta = \frac{c - Cb}{Sb}. \quad (12)$$

Dividing (10) by $U \sin \theta$ and applying (11)–(12) gives

$$2Ub = cF(R), \quad 2Uc = bF(D), \quad (13)$$

where

$$F(T) = V(T^2 - 2XT + 2) - ST. \quad (14)$$

Indeed, using $V = SX + CU$ and $U^2 + X^2 = 1$,

$$F(T) = VT^2 - (2CUX - 2SU^2 + 3S)T + 2SX + 2CU,$$

which is exactly the expression obtained from (10).

2. The circle computation

Place

$$A = (0, 0), \quad B = (c, 0), \quad C = (bC, bS).$$

Put $W = \cos(\alpha + x) = CX - SU$. The ray BK has direction $(-X, U)$ and the ray CL has direction $(-W, -V)$, so

$$K = \left(c - \frac{cRX}{2}, \frac{cRU}{2} \right), \quad (15)$$

$$L = \left(bC - \frac{bDW}{2}, bS - \frac{bDV}{2} \right). \quad (16)$$

Consequently,

$$|K|^2 = c^2 \left(1 - RX + \frac{R^2}{4} \right), \quad |L|^2 = b^2 \left(1 - DX + \frac{D^2}{4} \right). \quad (17)$$

Write the circumcircle of AKL as

$$\xi^2 + \eta^2 - P\xi - Q\eta = 0. \quad (18)$$

Its centre is $O = (P/2, Q/2)$. Set

$$\Delta = K_x L_y - K_y L_x.$$

Since the circumcentre is defined, A, K, L are noncollinear, hence $\Delta \neq 0$. Cramer's rule gives

$$P = \frac{P_0}{\Delta}, \quad Q = \frac{Q_0}{\Delta}, \quad (19)$$

where

$$P_0 = |K|^2 L_y - K_y |L|^2, \quad Q_0 = K_x |L|^2 - |K|^2 L_x. \quad (20)$$

Define

$$\mathcal{T} = 2(b(CP_0 + SQ_0) - cP_0) - (b^2 - c^2)\Delta. \quad (21)$$

Let

$$E_R = 2Ub - cF(R), \quad E_D = 2Uc - bF(D).$$

Substituting (14)–(17) and (20)–(21), and collecting terms, gives the polynomial identity

$$4V\mathcal{T} = A_0 E_R + B_0 E_D + (C^2 + S^2 - 1)G_0 + (X^2 + U^2 - 1)H_0, \quad (22)$$

where

$$A_0 = bc(C^2 D U b - C D U c + D S^2 U b - D S X c + 2 S c),$$

$$B_0 = bc(C R U b + R S X b - R U c - 2 S b),$$

$$G_0 = U b c^2 (2 C D R U X b - 2 C D U b + 2 D R S X^2 b - D R S b - 2 D S X b - 2 R U c),$$

$$H_0 = S b c (C D R U b^2 - C D R U c^2 + D R S X b^2 - D R S X c^2 - 2 D S b^2 + 2 R S c^2).$$

Identity (22) is verified by direct expansion of the displayed expressions.

By (13), $E_R = E_D = 0$, and $C^2 + S^2 = X^2 + U^2 = 1$. Also $V = \sin(\alpha + x) > 0$: indeed $0 < x < \gamma$, so $0 < \alpha + x < \pi$. Hence (22) gives $\mathcal{T} = 0$. By (19) and (21),

$$b(CP + SQ) - cP = \frac{b^2 - c^2}{2}. \quad (23)$$

Finally,

$$M = \left(\frac{c}{2}, 0\right), \quad N = \left(\frac{bC}{2}, \frac{bS}{2}\right).$$

Since $O = (P/2, Q/2)$,

$$\begin{aligned} OM^2 - ON^2 &= -\frac{cP}{2} + \frac{b(CP + SQ)}{2} + \frac{c^2 - b^2}{4} \\ &= 0 \end{aligned}$$

by (23). Therefore $OM = ON$.

IMO 2026 — Problem 3

Solution by GPT-5.6 Sol (max reasoning)

2026-07-22

Source: `results/gpt-5.6-sol-max/problem-03/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu Bang going first. Each player's goal is to maximise the total length of their own pieces. For each n , determine the largest value c such that Liu Bang may guarantee a total length of at least c , regardless of Xiang Yu's play.

Solution document

Status

unsolved

Problem

Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu Bang going first. Each player's goal is to maximise the total length of their own pieces. For each n , determine the largest value c such that Liu Bang may guarantee a total length of at least c , regardless of Xiang Yu's play.

Approaches tried

- Reformulated the claiming phase: rational play consists of taking a currently longest piece, so Liu Bang's payoff is the sum of the odd-indexed lengths after sorting all pieces nonincreasingly.

- Began investigating the equivalent alternating-sum discrepancy of refinements of Liu Bang's initial partition.

Current best

No final expression yet. If the final piece lengths are $x_1 \geq \dots \geq x_m$, Liu Bang receives $x_1 + x_3 + \dots$, equivalently $(1 + D)/2$ where $D = x_1 - x_2 + x_3 - \dots + (-1)^{m+1}x_m \geq 0$.

Full proof

Not yet complete.

IMO 2026 — Problem 4

Solution by GPT-5.6 Sol (max reasoning)

2026-07-22

Source: `results/gpt-5.6-sol-max/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ , then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

solved

Problem

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Approaches tried

- **Angle-coordinate game:** Normalize 180° to 1. Cutting from angle A and splitting it as $x + (A - x)$ in a triangle (A, B, C) yields children $(B, x, 1 - B - x)$ and $(C, A - x, B + x)$.

This exact transition model underlies both directions.

- **Finite rational-grid attractor computation:** For targets p/q and grid-restricted states/cuts through $q = 20$, all states were winning exactly when $p \mid q$, suggesting the answer $t = 1/n$. This was exploratory only; arbitrary real cuts required a proof independent of computation.
- **Reciprocal-target construction:** For $t = 1/n$, complementary angles at the cut point can be chosen as m/n and $(n - m)/n$. An interval argument finds an admissible m for every starting triangle; then an induction reduces any retained angle k/n to $1/n$.
- **Safe-family invariant for necessity:** For nonreciprocal t , call a triangle safe when no angle is a positive integral multiple of t . An exhaustive four-case use of the transition formula proves that every cut of a safe triangle has a safe child. An equilateral starting triangle is safe, so Shan-Yu can avoid the target forever.
- **Earlier acute-triangle idea:** For $t > 1/2$, retaining an acute child gives a special-case avoidance strategy, but the safe-family invariant supersedes it and handles all nonreciprocal targets uniformly.

Current best

The complete characterization is

$$\theta = \frac{180^\circ}{n} \quad \text{for some integer } n \geq 2.$$

For these and only these values, Mulan has a finite winning strategy. In fact, after at most one initialization cut she forces a positive integral multiple $k\theta$ in either possible child, and then needs at most $k - 1 \leq n - 2$ further cuts.

Full proof

Normalize all angle measures by dividing by 180° . Thus a triangle's angles sum to 1, and put

$$t = \frac{\theta}{180^\circ} \in (0, 1).$$

We prove that Mulan can force a win exactly when $t = 1/n$ for an integer $n \geq 2$.

1. The transition rule

Suppose the current triangle has angles (A, B, C) . Mulan cuts from the vertex of angle A to the opposite side, and let x be the portion of angle A lying in the child containing the vertex of angle B . Since the cut is interior,

$$0 < x < A.$$

The two children have angle triples

$$(B, x, 1 - B - x) \quad (1)$$

and

$$(C, A - x, B + x). \quad (2)$$

Indeed, the first two angles of the first child are B, x , so its third angle is $1 - B - x$. The second child has angles $C, A - x$ at old vertices, so its angle at the cut point is

$$1 - C - (A - x) = B + x.$$

Conversely, every $x \in (0, A)$ is attainable: the unique ray inside the angle A making angle x with the appropriate side meets the opposite side at an interior point.

2. Sufficiency when $t = 1/n$

Let $n \geq 2$ be an integer and suppose $t = 1/n$. We first show that, from any triangle not already containing angle t , Mulan can make a cut after which either possible child contains an angle k/n for some integer $k \in \{1, \dots, n-1\}$.

Let the triangle's angles be (A, B, C) .

If $n = 2$, then at least two angles are less than $1/2$: otherwise either one is exactly $1/2$, in which case Mulan has already won, or two are greater than $1/2$, which is impossible because the angle sum is 1. Label two angles below $1/2$ as B, C and the third as A . Then

$$C < \frac{1}{2} < 1 - B = A + C. \quad (3)$$

Now let $n \geq 3$, and choose A to be a largest angle. Then $A \geq 1/3$. In fact,

$$nA > 1. \quad (4)$$

For $n > 3$ this follows from $nA \geq n/3 > 1$. For $n = 3$, if $3A = 1$, then maximality and the angle sum force $A = B = C = 1/3 = t$, contrary to the assumption that the triangle does not already contain the target.

The open interval

$$(nC, n(A + C))$$

has length $nA > 1$, so it contains an integer m . To justify this explicitly, choose the least integer strictly greater than nC ; it is at most $nC + 1 < n(A + C)$. Moreover,

$$0 < nC < m < n(A + C) = n(1 - B) < n,$$

so $m \in \{1, \dots, n-1\}$. Consequently,

$$C < \frac{m}{n} < A + C. \quad (5)$$

For $n = 2$, the same conclusion holds with $m = 1$ by (3).

Mulan now cuts from the vertex with angle A , choosing

$$x = A + C - \frac{m}{n} = 1 - B - \frac{m}{n}.$$

The right inequality in (5) gives $x > 0$, and the left inequality gives $x < A$, so this is a legal cut. By (1) and (2), the angles at the new point in the two children are

$$1 - B - x = \frac{m}{n} \quad \text{and} \quad B + x = 1 - \frac{m}{n} = \frac{n - m}{n}. \quad (6)$$

Both are positive integral multiples of $t = 1/n$. Hence whichever child Shan-Yu retains has an angle kt with $1 \leq k \leq n-1$.

It remains to show that any such multiple can be reduced to the target. We use induction on k . If $k = 1$, Mulan has already won. If $k \geq 2$, she cuts from the vertex whose angle is kt , using the interior ray which splits it into

$$t + (k-1)t.$$

This is legal because both portions are positive. One child contains angle t , while the other contains angle $(k-1)t$. Thus Shan-Yu either retains a triangle with the target angle, or leaves a strictly smaller positive integral multiple, to which the induction hypothesis applies. Mulan therefore wins after finitely many cuts. This proves sufficiency.

3. Necessity for all other t

Suppose now that

$$t \neq \frac{1}{n} \quad \text{for every integer } n \geq 2. \quad (7)$$

Call a triangle **safe** if none of its angles is a positive integral multiple of t .

Shan-Yu can choose an equilateral initial triangle, and it is safe. Indeed, if its angle $1/3$ were kt for a positive integer k , then $t = 1/(3k)$, contrary to (7).

We claim that after any legal cut of a safe triangle, at least one child is safe. Let the safe parent have angles (A, B, C) , and suppose the cut is from the vertex of angle A . Use the parameter $x \in (0, A)$ from (1)–(2). Assume for contradiction that both children are unsafe.

The inherited angle B in the first child is not a positive integral multiple of t , because the parent is safe. Therefore one of the other two angles of that child must be such a multiple: for some positive integer r ,

$$x = rt \quad \text{or} \quad 1 - B - x = rt. \quad (8)$$

Likewise, inherited angle C in the second child is not a multiple, so for some positive integer s ,

$$A - x = st \quad \text{or} \quad B + x = st. \quad (9)$$

There are four possibilities.

- If $x = rt$ and $A - x = st$, then $A = (r+s)t$, contradicting safety of the parent.
- If $x = rt$ and $B + x = st$, then $B = (s-r)t$. As $B > 0$, we have $s-r \geq 1$, again contradicting safety.
- If $1 - B - x = rt$ and $A - x = st$, then

$$C = 1 - A - B = (1 - B - x) - (A - x) = (r-s)t.$$

Since $C > 0$, we have $r-s \geq 1$, contradicting safety.

- If $1 - B - x = rt$ and $B + x = st$, adding gives $1 = (r+s)t$. Hence $t = 1/(r+s)$. Since $r+s \geq 2$, this contradicts (7).

All four cases are impossible, proving the claim. Shan-Yu can therefore retain a safe child after every cut. Every retained triangle remains safe, and in particular never has an angle equal to t . Thus Mulan cannot guarantee a win when (7) holds.

Combining the two directions and restoring degrees gives exactly

$$\theta = \frac{180^\circ}{n} \quad (n = 2, 3, 4, \dots).$$

IMO 2026 — Problem 5

Solution by GPT-5.6 Sol (max reasoning)

2026-07-22

Source: results/gpt-5.6-sol-max/problem-05/current.md — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Approaches tried

- **Forward-orbit displacement (successful):** substituting $x = f(y)$ forces $f(f(y)) = 2f(y) - y$. Thus, for $d(t) = f(t) - t$, the displacement is constant along every forward orbit, whose terms are $t + nd(t)$. Positivity of all those terms gives $d(t) \geq 0$.
- **Asymptotic alignment of arithmetic orbits (successful):** for two points with positive displacements p, q , their forward orbits are arithmetic progressions. Terms of the first orbit can be aligned within a bounded distance of shifted terms of the second. The upper inequality then forces $p \leq q$; reversing the points forces $q \leq p$.
- **Connectedness after displacement quantization (successful):** once every positive displacement has one common value c , the sets where $d = 0$ and $d = c$ would both be open if both occurred. This contradicts connectedness of $\mathbb{R}_{>0}$.
- **Direct expansion of the squared inequalities (not used):** expanding produces polynomial bounds involving $d(x)$ and $d(y)$, but these are less transparent than rewriting the upper bound in terms of $x, f(y)$, and $d(x) - d(y)$.

- **Linear/translation ansatz (sanity check):** $f(x) = x + c$ works for every $c \geq 0$ by the AM–GM and RMS–AM inequalities; a homogeneous linear trial $f(x) = ax$ is consistent only for $a = 1$.
- **Numerical checks:** `code/sanity_check.py` tests the candidate family on random inputs across twelve orders of magnitude and checks the orbit-alignment estimate. Standalone write-ups of the three proof lemmas are in `lemmas/`.

Current best

The complete characterization is

$$\boxed{f(x) = x + c \quad \text{for every } x > 0, \quad c \geq 0.}$$

The proof first obtains nonnegative, forward-orbit-invariant displacement $d = f - \text{id}$; asymptotic alignment of two arithmetic forward orbits shows that all positive values of d coincide; finally, continuity of the numerical RMS–AM gap and connectedness of $\mathbb{R}_{>0}$ rule out coexistence of displacement 0 with a positive displacement.

Full proof

Put

$$d(t) = f(t) - t \quad (t > 0).$$

1. Forward orbits and nonnegative displacement

(Standalone version: [orbit-structure lemma](#).) Fix $y > 0$ and substitute $x = f(y)$. The two outer expressions in the given chain are then both $f(y)$, since

$$\sqrt{\frac{f(y)^2 + f(y)^2}{2}} = f(y) \quad \text{and} \quad \sqrt{f(y)f(y)} = f(y).$$

Consequently equality holds throughout, and hence

$$f(f(y)) = 2f(y) - y. \quad (1)$$

It follows that

$$d(f(y)) = f(f(y)) - f(y) = f(y) - y = d(y). \quad (2)$$

For $n \geq 0$, let f^n denote the n -fold iterate (with f^0 the identity). Equations (1)–(2), followed by induction, give

$$d(f^n(y)) = d(y), \quad f^n(y) = y + nd(y) \quad (n \geq 0). \quad (3)$$

Every $f^n(y)$ is positive. If $d(y) < 0$, however, then the second formula in (3) is negative for every sufficiently large integer n , a contradiction. Therefore

$$d(y) \geq 0 \quad \text{for all } y > 0. \quad (4)$$

2. All positive displacements are equal

(Standalone version: [positive-displacements-equal lemma](#).) Let $a, b > 0$ have positive displacements

$$p = d(a) > 0, \quad q = d(b) > 0.$$

By (3), for all nonnegative integers m, n ,

$$x := f^m(a) = a + mp, \quad y := f^n(b) = b + nq,$$

and $d(x) = p$, $d(y) = q$. Write

$$v = f(y) = b + (n+1)q.$$

The left-hand inequality of the problem (that is, the upper bound on the middle expression) becomes

$$\sqrt{\frac{x^2 + v^2}{2}} \geq \frac{f(x) + y}{2} = \frac{(x+p) + (v-q)}{2}.$$

After multiplication by 2, this yields

$$p - q \leq \Phi(x, v), \quad \Phi(u, v) := \sqrt{2(u^2 + v^2)} - u - v. \quad (5)$$

For every sufficiently large m , choose

$$n = n_m := \left\lfloor \frac{a + mp - (b + q)}{q} \right\rfloor.$$

Then $n_m \geq 0$, and with $x_m = a + mp$ and $v_m = b + (n_m + 1)q$ we have

$$0 \leq x_m - v_m < q. \quad (6)$$

Moreover, $x_m, v_m \rightarrow \infty$. Rationalizing gives, for all positive u, v ,

$$\Phi(u, v) = \frac{(u - v)^2}{\sqrt{2(u^2 + v^2)} + u + v}. \quad (7)$$

Thus (6)–(7) imply

$$0 \leq \Phi(x_m, v_m) \leq \frac{q^2}{x_m + v_m} \rightarrow 0.$$

Applying (5) along this sequence gives $p - q \leq 0$. Interchanging a and b and repeating the same argument gives $q - p \leq 0$. Hence

$$p = q. \quad (8)$$

Therefore, either d vanishes identically, or there is a number $c > 0$ such that

$$d(t) \in \{0, c\} \quad (t > 0). \quad (9)$$

3. The two displacement values cannot coexist

(Standalone version: [no-mixed-displacement lemma](#).) Suppose now that $c > 0$ occurs, and define

$$A = \{t > 0 : d(t) = 0\}, \quad B = \{t > 0 : d(t) = c\}.$$

By (9), these sets are disjoint and cover $\mathbb{R}_{>0}$, and B is nonempty.

If $x \in B$ and $y \in A$, then $f(x) = x + c$ and $f(y) = y$. The left-hand inequality in the problem gives

$$\sqrt{\frac{x^2 + y^2}{2}} \geq \frac{x + c + y}{2},$$

or equivalently

$$c \leq \Phi(x, y), \quad (10)$$

where Φ is as in (5). The function Φ is continuous on $\mathbb{R}_{>0}^2$ and satisfies

$$\Phi(t, t) = 0 \quad (t > 0). \quad (11)$$

Fix $b \in B$. By (11) and continuity, all y sufficiently close to b satisfy $\Phi(b, y) < c$. Such a y cannot belong to A , by (10), so it belongs to B . Hence B is open in $\mathbb{R}_{>0}$. Similarly, if $a \in A$, then all x sufficiently close to a satisfy $\Phi(x, a) < c$; (10) shows that such an x cannot belong to B . Hence A is also open in $\mathbb{R}_{>0}$.

The interval $\mathbb{R}_{>0}$ is connected, so it cannot be the union of two disjoint nonempty open subsets. Since B is nonempty, we conclude that A is empty. Thus $d(t) = c$ for every $t > 0$. Together with the case $d \equiv 0$, this proves that every solution must have the form

$$f(t) = t + c \quad (t > 0)$$

for some $c \geq 0$.

4. Verification

Conversely, fix $c \geq 0$ and let $f(t) = t + c$. For all $x, y > 0$, set $z = y + c > 0$. Then

$$\frac{f(x) + y}{2} = \frac{x + z}{2}.$$

The RMS–AM and AM–GM inequalities applied to the positive numbers x, z give

$$\sqrt{\frac{x^2 + z^2}{2}} \geq \frac{x + z}{2} \geq \sqrt{xz}.$$

Since $z = f(y)$, this is precisely the required chain. Hence every $c \geq 0$ indeed gives a solution, completing the characterization.

IMO 2026 — Problem 6

Solution by GPT-5.6 Sol (max reasoning)

2026-07-22

Source: `results/gpt-5.6-sol-max/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n , the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \dots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n . (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y .)

Solution document

Status

partial

Problem

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n , the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \dots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n . (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y .)

Approaches tried

- Recast the condition using prime-divisor sets. Define $S_n = \{x > 1 : \gcd(x, a_i) > 1 \text{ for all } i \leq n\}$. Then $S_{n+1} \subseteq S_n$, and a_{n+1} is the least member of S_n above a_n .
- Identified a sufficient stabilization criterion: if S_N is pairwise non-coprime, then every later term lies in S_N and imposes no new restriction on it, so $S_n = S_N$ thereafter. The sequence is then the increasing enumeration, starting with a_1 , of a periodic set of integers.
- Established the bounded-gap observation: every multiple of a_1 belongs to every S_n , hence $1 \leq a_{n+1} - a_n \leq a_1$.

- Explored finite prime supports and computed examples. For $a_1 = 15$, the sequence starts 15, 18, 20, 24, 30, 36, 40, 42, 45, 48, 50, 54, 60, ... and has $(T, L) = (8, 30)$. After the first three supports $\{3, 5\}, \{2, 3\}, \{2, 5\}$, the stable admissible set is the set of numbers divisible by at least two of 2, 3, 5.
- The simple hope that a stable admissible set must be a single divisibility class is false; the $a_1 = 15$ example shows that a finite union of classes is needed.

Current best

It remains to prove finite stabilization. The problem has been reduced to a combinatorial/arithmetic assertion about the descending admissible sets S_n . Each S_n is periodic (its membership depends only on divisibility by the finitely many primes occurring in $a_1, \dots, a_n$), but new primes may enter later terms. A likely route is to use the bounded gaps together with the fact that every term has a prime factor from the fixed finite set $P(a_1)$, or to formulate the process as successively adjoining hyperedges and prove that it reaches a pairwise-intersecting blocker after finitely many steps. Once stabilization is proved, periodicity of the sequence, including the identity from $n = 1$, can be derived by choosing a common period divisible by all relevant primes and checking the counting shift.

Full proof

Not yet complete. Established pieces are as follows.

Let

$$S_n = \{x > 1 : \gcd(x, a_i) > 1 \quad (1 \leq i \leq n)\}.$$

Every multiple of a_1 lies in S_n , since each a_i has gcd greater than 1 with a_1 . Thus the recurrence is always defined and

$$a_n < a_{n+1} \leq a_n + a_1.$$

If for some N every two members of S_N have gcd greater than 1, then all future terms belong to S_N . Each such term is non-coprime to every member of S_N , so adding it as a constraint does not shrink S_N . Hence $S_n = S_N$ for all $n \geq N$, and the recurrence successively lists every member of S_N above the current term. Since S_N is defined by finitely many gcd conditions, it is periodic modulo the product of the finitely many primes dividing $a_1 \cdots a_N$. This gives a constant index shift under translation by that modulus. The remaining issue is to prove that such an N exists and then make the all- n indexing argument fully explicit.